

# BOGYEOM PARK

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## RESEARCH INTEREST

I design and study AI-supported learning systems that promote deeper cognitive engagement, at the intersection of AI in Education, Human-AI Interaction, and Learning Analytics.

- Eliciting deeper engagement - designing AI tutors that prompt learners to explain, compare, reflect, and build on dialogue rather than receive answers passively
- Understanding learning processes - analyzing conversational and behavioral traces to characterize cognitive engagement and connect interaction patterns with learning outcomes

## EDUCATION

**Seoul National University of Science and Technology**  
**Integrated Ph.D. in Applied Artificial Intelligence**

- Advisor: Kyoungwon Seo

Seoul, South Korea  
Mar. 2023 - Present

**Seoul National University of Science and Technology**  
**B.S. in Electrical and Information Engineering**

- GPA: 4.14/4.5 (Major GPA: 4.5/4.5)

- Thesis: Online Learning Support System Based on Facial Recognition

Seoul, South Korea  
Mar. 2019 - Feb. 2023

## HONORS & AWARDS

- AI SeoulTech Graduate Scholarship (KRW 10,000,000), Seoul Scholarship Foundation, 2025
- Best Student Paper Award, IEEE Seoul Section, 2024
- National Science and Engineering Undergraduate Scholarship (full tuition), Korea Student Aid Foundation, 2021-2022
- Best Capstone Design Award, Seoul National University of Science and Technology, 2022

## REFEREED JOURNAL ARTICLE

**Integrating Biomarkers From Virtual Reality and Magnetic Resonance Imaging for the Early Detection of Mild Cognitive Impairment Using a Multimodal Learning Approach: Validation Study**

Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo

Journal of Medical Internet Research, 26, e54538 (2024) - SCIE; JCR Top 3%; Q1

## EXTENDED ABSTRACTS

**Assessing Critical Thinking Through a Multi-Agent LLM-Based Debate Chatbot**

Bogyeom Park and Kyoungwon Seo

CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems

**Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment**

Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo

CHI EA '24: Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems

## RESEARCH EXPERIENCE

**Human-centered Artificial Intelligence (HAI) Lab, SeoulTech**

**Graduate Researcher (Advisor: Kyoungwon Seo)**

**ICAP-Based AI Tutoring System for Probability and Statistics Learning**

**Lead Researcher | Korea Education & Research Information Service**

Seoul, South Korea  
Mar. 2023 - Present

Mar. 2026 - Aug. 2026

- Designed and evaluated an ICAP-based AI tutor that uses staged elicitation to promote active, constructive, and interactive engagement rather than answer delivery
- Developed an utterance-level coding and analytics workflow linking dialogue evidence with tutor correctness, usage patterns, and learning outcomes
- Built an interactive research dashboard for reviewing engagement labels and comparing AI-tutored learning processes

## GUI Agent Technologies for Automated UX Accessibility Evaluation

### Research Assistant | National Research Foundation of Korea

Sep. 2025 - Present

- Designed a GUI agent capable of performing expert-level automated evaluations of UX accessibility
- Planned experimental protocols for validating human-AI comparative performance in accessibility assessments

## Agentic AI for Personalized Career, Academic, and Counseling Support

### Lead Researcher | Korean Educational Development Institute

Jun. 2025 - Dec. 2025

- Investigated how agentic AI could support personalized educational pathways, counseling, and decision-making

## AI Copilot Technologies for Adaptive, Teacher-Augmented Learning

### Research Assistant | Institute for Information & Communications Technology

Jul. 2023 - Aug. 2025

#### Planning & Evaluation

- Built counseling chatbot and analysis models supporting AI-driven student coaching
- Co-developed an integrated platform enabling personalized teacher-augmented learning

## Human-centered Artificial Intelligence (HAI) Lab, SeoulTech

Seoul, South Korea

### Undergraduate Researcher (Advisor: Kyoungwon Seo)

Jul. 2021 - Feb. 2023

## LLMs to Support Teachers in Educational Settings

### Research Assistant | Lab Project

Mar. 2023 - Feb. 2024

- Fine-tuned an LLM to generate student competency-analysis reports and assessed its usefulness through expert interviews
- Identified opportunities and limitations of LLM support for competency assessment and report generation

## Multimodal Digital Biomarkers for Early Dementia Diagnosis

### Research Assistant | National Research Foundation of Korea

Mar. 2022 - Feb. 2024

- Collected VR kiosk interaction data from participants with mild cognitive impairment and healthy controls in collaboration with Hanyang University Guri Hospital
- Developed multimodal predictive models integrating VR behavioral and MRI biomarkers for early detection of mild cognitive impairment

## TEACHING

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### Deep Learning

Fall 2023

Teaching Assistant, Seoul National University of Science and Technology

- Designed a final project using CNN-based models to predict drivers' physical and cognitive states from image data
- Supported lectures, advised student projects, and graded assignments and examinations

## SKILLS

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- **Research Methods** - experimental design, usability evaluation, user research, prototyping, multimodal learning analytics, statistical analysis, qualitative coding
- **AI and Data** - machine learning, deep learning, feature engineering, LLM applications, AI agents
- **Programming and Analysis** - Python, C/C#, SPSS
- **Design** - HCI theory, UX design, interaction prototyping, usability testing

## PATENT

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Explainable AI-Based System for Early Diagnosis and Prognosis Prediction of Alzheimer's Disease Using VR Biomarkers, and Method Thereof (Korean Patent Application No. 10-2023-0105821, filed Aug. 2023)

## SERVICE

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- Publicity Manager, SeoulTech Human-centered Artificial Intelligence Lab